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Identifying the needs of low-income angioplasty patients and caregivers to inform the design of a support platform

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ABSTRACT: This research investigates the needs and preferences of low-income angioplasty patients and their caregivers in India during post-angioplasty recovery. Through in-depth interviews and contextual inquiries, the study uncovers critical informational, physical, and emotional needs. Patients often lack access to reliable health information, leading to misconceptions about care and medication adherence. Pain management and emotional support are significant concerns for both patients and caregivers. The study proposes the integration of digital health solutions to address these challenges, providing a platform for reliable information, communication, and support. This research emphasizes the need for context-sensitive interventions to improve patient outcomes and enhance the quality of life for vulnerable populations in developing countries.

KEYWORDS: artificial intelligence, user centred design, service design, post-angioplasty care, low income patient

1. Introduction

Coronary angioplasty, a minimally invasive procedure for treating coronary artery disease, has become a cornerstone of cardiac care, particularly for patients experiencing acute coronary events. As the prevalence of cardiovascular diseases continues to rise globally, improving post-angioplasty recovery through effective support systems is crucial, especially among low-income populations who face additional barriers to accessing healthcare resources.

Existing research highlights that effective postoperative support significantly influences patient adherence to treatment plans and overall recovery (Hasankhani et al., 2014). However, many patients struggle with pain management, medication adherence, and lifestyle modifications during their recovery phase (Ibrahim et al., 2022). While previous studies have examined general recovery challenges, they largely overlook how financial constraints shape these experiences, particularly in developing countries with uneven healthcare access (Subrahmanian et al., 2020). Furthermore, caregivers—often family members—play a critical role in supporting patients, yet their needs and preferences remain underexplored in the context of digital healthcare solutions.

The gap in research is not only about understanding the needs and preferences of low-income angioplasty patients and caregivers but also about identifying design opportunities for interventions tailored to this population. Current digital health solutions often assume a one-size-fits-all approach, failing to address the unique constraints faced by low-income patients, such as limited digital literacy, financial barriers, and reliance on informal information sources. There is limited knowledge on how to design effective digital support systems that provide credible information, facilitate symptom tracking, enhance medication adherence, and integrate emotional support in an accessible and cost-effective manner.By employing qualitative methods such as semi-structured interviews and contextual inquiries, this research seeks to uncover the key requirements and barriers for designing

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effective, context-sensitive design solutions. The findings will contribute to the existing body of knowledge by providing insights into how design solutions can be tailored to address the unique needs of this vulnerable population. Ultimately, this study will inform the development of design strategies that enhance recovery experiences and improve health outcomes for low-income patients and their caregivers in post-angioplasty care.

2. Literature review

The rising prevalence of Coronary Artery Disease (CAD) has made coronary angioplasty, or Percutaneous Coronary Intervention (PCI), an essential cardiovascular procedure. Angioplasty primarily helps relieve angina symptoms and restore blood flow to the heart, reducing the risk of myocardial infarction. While the procedure is effective, postoperative recovery presents significant challenges, particularly for low-income patients and their caregivers. Despite substantial research on the postoperative needs of angioplasty patients, there is limited literature addressing the unique challenges faced by low-income individuals, making this an essential area for further investigation.

2.1. Known postoperative needs and challenges for angioplasty patients

Existing studies indicate that inadequate pain management and lack of emotional support contribute to chronic pain, anxiety, and depression, leading to poor health outcomes (Ibrahim et al., 2022). Patients also need long-term adherence to medication and lifestyle changes to prevent restenosis and improve survival rates (Astin et al., 2008). However, low-income patients face additional socioeconomic barriers, including financial constraints, limited health literacy, transportation challenges, and inadequate follow-up care (Rathore et al., 2006).

A major concern is medication adherence, as high out-of-pocket costs for essential medications such as antiplatelet therapy and statins often lead to non-compliance, increasing the risk of complications and hospital readmissions (Kripalani et al., 2007). Similarly, dietary modifications crucial for cardiac health are difficult to maintain due to financial limitations and poor access to nutritious food options in low-income communities.

Cardiac rehabilitation programs are essential for reducing future cardiac events and improving recovery. However, low-income patients are significantly less likely to participate due to financial constraints, lack of awareness, and geographic barriers (Clark et al., 2005). Many rely on government or charitable healthcare services, which often provide inconsistent post-procedure monitoring and preventive care.

Caregivers are vital to angioplasty patients' recovery, influencing medication adherence and lifestyle changes. However, their needs are often overlooked. Low-income caregivers face heightened challenges due to financial strain, limited health knowledge, and lack of support, impacting both their well-being and patient outcomes (Vashistha et al., 2019).

2.2. Research gap in existing studies

While extensive research has explored post-angioplasty challenges, most studies focus on the general patient population rather than low-income individuals. Additionally, research on low-income cardiovascular patients often includes broader heart disease populations rather than angioplasty-specific challenges. As a result, there is no dedicated literature examining the unique needs and barriers faced by low-income angioplasty patients and their caregivers. This study explores post-angioplasty challenges for low-income patients and caregivers, identifying key pain points to inform user-centered digital health solutions.

2.3. Qualitative approaches to identifying barriers in healthcare

Qualitative research employs various data analysis methods to identify needs, challenges, and barriers in healthcare. Grounded Theory (Glaser & Strauss, 2017) helps develop theories from data but is time-intensive for applied healthcare studies (Walker & Myrick, 2006). Phenomenological Analysis (Smith & Osborn, 2004) captures lived experiences but may not address broader systemic challenges. Content Analysis (Elo & Kyngäs, 2008) systematically quantifies qualitative data but often lacks depth. Framework Analysis (Gale et al., 2013) offers structure for policy research but can be rigid. In contrast, Thematic Analysis (Braun & Clarke, 2006) provides flexibility and depth, making it ideal for exploring barriers in post-angioplasty care while allowing practical application in healthcare design solutions.

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2.4. Digital health solutions for post-angioplasty care

The integration of digital health solutions has emerged as a promising approach to address the complex needs of angioplasty patients and caregivers. Digital therapeutics have shown significant improvements in patient outcomes by enhancing engagement, adherence, and self-management (Srinivasan, 2023). Studies highlight that digital platforms can improve clinical parameters such as blood pressure and heart rate following cardiac events (Srinivasan, 2023). These innovations hold potential for reducing disparities in post-angioplasty care, particularly for low-income patients with limited access to healthcare facilities.

3. Methodology

3.1. Qualitative study

This qualitative study utilized an ethnographic approach to explore angioplasty patients' and their caregivers' needs and preferences during the postoperative recovery phase. Data collection methods included contextual inquiries in the angioplasty setting at two public hospitals, as well as semi-structured interviews with patients and caregivers in India.

3.2. Participant selection

Participants were selected based on specific criteria to ensure relevance. Included were angioplasty patients from public hospitals, their accompanying caregivers, and individuals aged 18+ from low-income backgrounds. Patients with cognitive impairments affecting consent or communication were excluded. The study involved 10 patients and 21 caregivers, with their characteristics detailed in Tables 1 and 2.

Table 1. Demographics of patients

Patient ID	Hospital	Gender	Age	Angioplasty Type	Education	Income Per Month
P1	Public	Male	46	Emergency	B.Com	₹ 25000
P2	Public	Female	51	Elective	Secondary (SSC)	₹ 15000
P3	Public	Male	64	Emergency	Primary (Up to V)	₹ 20000
P4	Public	Male	46	Elective	BA	₹ 30000
P5	Public	Male	53	Emergency	Primary (Up to IX)	₹ 13000
P6	Public	Female	59	Emergency	Primary (Up to VIII)	₹ 16000
P7	Public	Female	56	Emergency	Secondary (SSC)	₹ 20000
P8	Public	Male	51	Emergency	HSC	₹ 10000
P9	Public	Male	62	Emergency	HSC	₹ 22000
P10	Public	Male	65	Emergency	Secondary (SSC)	₹ 12000

Table 2. Demographics of caregivers

Characteristics	Total (N=21)	Male (N=16)	Female (N=5)
Age	25–30: 6	4	2
	30–35: 7	4	3
	35–40: 8	7	1
Relation ship	Son: 16	16	0
	Wife: 4	0	4
	Daughter: 2	0	2
Education	Graduate: 17	14	3
	Higher Secondary: 4	2	2
Employment	Unemployed: 8	5	3
	Employed: 13	11	2
Monthly Income	<₹10K: 5	4	1
	₹10K–₹15K: 6	4	2
	₹20K–₹25K: 2	2	0

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3.3. Data collection

Data were collected through semi-structured interviews and contextual inquiries to explore the postoperative experiences of angioplasty patients and their caregivers. Interviews lasted 45–60 minutes, and informed consent was obtained from all participants. The interview guide was developed based on a literature review of postoperative recovery, caregiver burden, and patient-centered care. Key themes such as pain management, emotional support, medication adherence, and caregiver challenges shaped the interview structure. Open-ended questions encouraged detailed responses, with follow-up prompts used for deeper exploration. The interview guide was tailored for patients and caregivers to reflect their unique experiences. Patients were first interviewed in the presence of their caregivers for emotional support, followed by separate caregiver interviews. Patient interviews focused on recovery, support systems, and medication adherence, while caregiver interviews explored their responsibilities, emotional challenges, and interactions with healthcare providers.

Contextual inquiries were conducted over a week in angioplasty wards at two public hospitals, lasting 2–3 hours per session. Observations focused on patient-caregiver interactions, emotional responses, and environmental factors affecting recovery. A structured template documented key aspects such as communication with healthcare staff, caregiver support dynamics, and accessibility barriers.

Interviews were conducted in Hindi and transcribed into English. Twelve interviews were recorded, while for others, detailed notes were taken due to patient preferences. Key observations from contextual inquiries were documented using sticky notes for reference. This combined approach provided in-depth qualitative insights into the postoperative needs of low-income patients.

3.4. Data analysis

The collected data from semi-structured interviews and contextual inquiries were analyzed using thematic analysis, following the six-phase framework proposed by Braun and Clarke (2006). This qualitative approach allowed for identifying patterns within the data, capturing the lived experiences of low-income angioplasty patients and their caregivers.

An inductive approach was used, allowing themes to emerge directly from the data rather than being shaped by preconceived hypotheses. Data from patients and caregivers were analyzed separately to ensure their unique perspectives were fully captured. Each transcript was coded individually, followed by a comparative analysis to identify shared and contrasting experiences.

The analysis process began with familiarization, where interview recordings were transcribed verbatim, and notes from contextual inquiries were compiled. These transcripts were read multiple times to gain a deep understanding of the content. Initial coding was conducted by identifying meaningful data segments and assigning descriptive labels. For instance, a patient expressing concern about their future health was coded under “Fear of Recurrence.” After generating initial codes, related codes were grouped into broader themes. For example, concerns about heart health, uncertainty about future complications, and anxiety regarding lifestyle changes were categorized under “Emotional Anxiety.” The themes were then reviewed and refined to ensure they accurately represented participants' experiences. Adjustments were made where necessary to capture nuances in their narratives. Each theme was clearly defined and named to maintain clarity and alignment with research objectives.

For example, a patient stated, “I worry about the possibility of heart attacks and overall health.” This was initially coded as “Fear of Recurrence.” As similar concerns emerged across multiple participants, this code evolved into a sub-theme of “Emotional Anxiety.” Further connections were found between Uncertainty about Future Health and Lack of Emotional Support, leading to the overarching theme of “Unmet Emotional Needs.” This broader theme encapsulated the significant psychological burden experienced by both patients and caregivers during recovery. The final themes provided a comprehensive understanding of the physical, informational, and emotional challenges faced by participants. This analysis offered valuable insights for designing interventions that address their specific needs, ensuring a more supportive and effective post-angioplasty recovery process.

4. Results

The findings from this study reveal critical insights into the needs and preferences of angioplasty patients and their caregivers during the postoperative recovery phase. These findings are categorized into informational, physical, and emotional needs. These categories were derived from a synthesis of existing models in the domain, including the Information-Motivation-Behavioural Skills (IMB) model for

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understanding health behaviours and Engel's model for patient behaviour post intervention. These models were chosen for their well-established use in understanding the multifaceted needs of patients in healthcare contexts and adapted to capture the specific challenges faced by angioplasty patients and their caregivers.

4.1. Informational needs

The findings related to informational needs highlight significant challenges faced by angioplasty patients and their caregivers in accessing reliable health information. Many participants reported relying on non-medical sources, such as social media and anecdotal advice, which often led to misconceptions about postoperative care. The need for clear guidelines on do's and don'ts during recovery emerged as a critical theme, with patients expressing a desire for tailored advice on lifestyle modifications, symptom management, and medication adherence. Furthermore, the importance of effective communication with healthcare providers was emphasized, as patients sought immediate access to clarify doubts and receive credible information. Table 3 presents a brief overview of the specific informational needs.

Table 3. Informational needs of participants post angioplasty

Challenges	Description	Need Identified	Quotes from Participants
Diverse Information Sources Impact Decision-Making **(Reported by 31 out of 31 participants)**	Patients and caregivers often rely on social media and anecdotal advice for health-related information, leading to misconceptions.	Need for credible and curated information to guide decisions.	*"Feeling tired of medication, I came across Social media information suggesting herbal supplements could cure heart problems."*
Need for Do's and Don'ts for New Situations **(Reported by 29 out of 31 patients)**	Patients require guidance on everyday habits, dietary habits, and exercise when new situations arise.	Clear guidelines on lifestyle modifications and activities.	*"I want to know what I can and cannot do when I face new symptoms or situations."*
Clarification on Unexpected Symptoms **(Reported by all participants)**	Patients and caregivers want to know if unexpected symptoms are typical or serious and when to seek help.	Access to healthcare professionals for symptom evaluation.	*"When I see unexpected symptoms, I want to know if they are typical or serious."*
Addressing Doubts and Questions **(Reported by 23 out of 31 participants)**	Immediate access to healthcare professionals for any sudden doubts or concerns that arise during recovery is essential.	Quick communication channels with healthcare providers.	*"When doubts suddenly come to my mind, I want to reach out to someone who can clarify them."*
Understanding Medication **(Reported by 19 out of 31 participants)**	Patients need detailed information about medications, including side effects, dosage, and risks of missing doses.	Comprehensive medication education resources.	*"I need to learn more about my medications and what happens if I miss a dose."*
Medication Changes Guidance **(Reported by 27 out of 31 participants)**	Patients and caregivers require information on how to make changes to medication if issues arise post-surgery.	Support in managing medication adjustments safely.	*"If there are problems with my medication, I want to know how to make changes safely."*
Regular Follow-ups and Check-ups **(Reported by 24 out of 31 participants)**	Clear instructions regarding the schedule and importance of follow-up appointments after surgery are needed.	Understanding of follow-up care requirements.	*"I want to understand the importance of regular follow-ups and what they entail."*

4.2. Physical needs

The findings regarding physical needs reveal the multifaceted challenges that patients encounter during the postoperative recovery phase. Pain management emerged as a prominent concern, with many patients struggling to find effective relief without becoming dependent on pain medications. Additionally,

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adherence to complex medication regimens proved difficult for some participants, highlighting the need for simplified management tools and reminders. Lifestyle modifications, including dietary changes and increased physical activity, were also identified as significant challenges. Table 4 summarizes the detailed physical needs of the patients.

Table 4. Physical needs of participants post angioplasty

Physical Challenges	Description	Need Identified	Quotes from Participants
Pain Management Issues **(Reported by 08 out of 10 patients)**	Patients experience significant pain after angioplasty, making it difficult to manage without becoming dependent on painkillers	Effective pain management strategies and support resources.	*"The pain after angioplasty was unbearable at times, and I struggled to find relief without becoming dependent on painkillers."*
Medication Adherence Difficulties **(Reported by 06 out of 10 patients)**	There is a struggle with adhering to complex medication regimens due to forgetfulness or lack of understanding.	Simplified medication management tools and reminders.	*"Sometimes I forget to take my medication because I'm not fully aware of potential complications post angioplasty."*
Lifestyle Modifications Struggles **(Reported by 10 out of 10 patients)**	Patients find it challenging to adhere to recommended dietary changes and lifestyle modifications post-surgery.	Guidance on lifestyle changes tailored for recovery.	*"Concerned about dietary changes and struggling with quitting smoking as part of the recommended lifestyle modifications."*
Physical Discomfort and Limitations **(Reported by 10 out of 10 patients)**	Many patients experience discomfort at the catheter insertion site or other physical limitations that hinder recovery.	Information on managing physical discomfort effectively	*"I feel pain in the groin or at the puncture site from where the catheter was inserted during angioplasty."*

4.3. Emotional needs

The emotional needs identified in the study highlight the psychological challenges faced by angioplasty patients and their caregivers during recovery. Many participants reported feelings of isolation and anxiety related to their health status, particularly concerning the fear of future cardiovascular events. Participants expressed a strong desire for community support and reassurance from healthcare providers to alleviate their fears and enhance their emotional well-being. These findings suggest that integrating emotional support resources into post-angioplasty care is essential for fostering resilience and improving overall recovery experiences for both patients and caregivers. Table 5 presents a detailed breakdown of the emotional needs expressed by participants.

Table 5. Emotional needs of participants post angioplasty

Emotional Challenges	Description	Need Identified	Quotes from Participants
Feelings of Isolation and Misunderstanding **(Reported by 07 out of 10 patients)**	Patients often feel isolated in their recovery journey, struggling with feelings of being misunderstood by healthcare providers.	Emotional support systems that foster connection.	*"I often feel isolated and misunderstood by healthcare providers and loved ones."*
Uncertainty and Fear About Recovery **(Reported by 08 out of 10 patients)**	Many patients experience anxiety regarding their health status post-surgery, fearing complications or setbacks in recovery.	Reassurance and clear communication from healthcare providers.	*"I worry about the possibility of heart attacks and overall health."*

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5. Discussion

This study highlights the informational, physical, and emotional needs of low-income angioplasty patients and their caregivers, addressing a critical gap in existing research. While prior studies have explored general post-angioplasty recovery, few focus on the unique challenges faced by low-income populations, making this research essential for designing targeted support solutions.

Existing literature underscores the importance of patient education and access to reliable health information in improving post-angioplasty outcomes (Asitin et al., 2008). However, this study highlights the specific challenges faced by low-income individuals, who often lack direct contact with healthcare providers post-discharge, leading them to seek information from alternative, non-medical sources (Bashshur et al., 2016). Our findings align with research indicating that patients frequently encounter conflicting information from various sources, leading to confusion regarding their care and treatment plans (Jin et al., 2025). However, our research extends this knowledge by demonstrating how financial constraints exacerbate the issue, as low-income patients may be unable to afford consultations with healthcare professionals for minor doubts, thus increasing their reliance on unreliable sources.

This research confirms that pain management and lifestyle modifications, already known to aid angioplasty recovery (Ibrahim et al., 2022), are particularly challenging for low-income patients. Limited access to effective strategies and complex medication adherence, driven by financial constraints (Kripalani et al., 2007), necessitate tailored, affordable interventions, and this point the previous research didn't address.

Emotional support is crucial for angioplasty patients (Ibrahim et al., 2022), but this study highlights the added stress and uncertainty faced by their caregivers. Integrating emotional support resources for both patients and caregivers, especially with a focus on the financial needs of the patient, and with guidance from the psychological needs of patients and their caregivers, becomes a key priority.

5.1. Addressing the unique needs of low-income angioplasty patients

Our study highlights the unique challenges faced by low-income angioplasty patients, a group often overlooked in digital health interventions. While all patients require medical information, emotional support, and lifestyle guidance, financial and logistical barriers shape how these needs manifest for low-income individuals. Unlike higher-income patients who can directly consult cardiologists, low-income patients in government hospitals rely on lengthy in-person visits, delaying critical guidance.

Medication adherence is also impacted by affordability, with some patients discontinuing treatment prematurely or opting for cheaper, unverified alternatives, increasing health risks. Additionally, emotional support is deeply tied to family networks, as low-income patients lack access to professional psychological support. These distinct challenges call for a tailored digital health intervention that provides curated, credible, and accessible medical information, structured recovery guidance, and emotional support tools. By addressing these gaps, such solutions can improve adherence, reduce misinformation, and enhance post-angioplasty care for underserved populations.

5.2. Application of results: design opportunities for digital care applications

The insights gained from patient and caregiver interviews underscore the necessity for a tailored digital platform that provides relevant and personalized information. The identified informational needs highlight several key areas where a digital solution can make a significant impact.

5.2.1. Addressing misinformation through an AI-based health platform

Our study found that many patients rely on non-medical sources due to limited healthcare access and financial constraints. This often leads to misinformation, confusion, and harmful self-management practices.

Design opportunity

How might we provide credible, accessible, and personalized health information on lifestyle changes, symptom management, medication adherence, and diet?

We propose an AI-driven platform that provides personalized health insights by considering a patient's medical history, preferences, cultural background, and financial constraints. It offers symptom-based guidance, ensuring timely, reliable feedback aligned with medical guidelines. Additionally, culturally and budget-friendly dietary recommendations enhance accessibility. By leveraging AI, this solution reduces misinformation and empowers patients with tailored, credible health support.

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5.2.2. Reducing anxiety through AI-driven symptom tracking

Our study highlights significant anxiety among patients regarding symptom interpretation. Many participants expressed uncertainty about when to seek medical attention, leading to unnecessary worry or delayed care.

Design opportunity

How might we enable patients and caregivers to track symptoms, receive real-time feedback, and distinguish between typical post-operative experiences and potential complications?

We propose an AI-powered chatbot as the first point of contact for symptom assessment, using evidence-based protocols to provide self-care guidance and flag emergencies. Visual aids like charts and graphs help patients track symptom trends, enhancing their understanding and control. This approach reduces anxiety, improves condition awareness, and ensures appropriate healthcare utilization.

5.2.3. Enabling easy access to healthcare guidance

Our study found that patients often seek immediate clarification for their doubts and concerns but face barriers in accessing healthcare professionals due to cost or logistical challenges.

Design opportunity

How might we establish secure, cost-effective, and user-friendly channels that enable patients to connect with healthcare providers easily, fostering trust and encouraging engagement with treatment plans?

To address this, we propose a service enabling government healthcare officials to provide timely support through a secure messaging system, allowing patients to seek medical guidance within a set timeframe. A regularly updated FAQ section would address common concerns with credible information. This approach enhances healthcare accessibility, reduces misinformation, and builds patient confidence in their treatment journey.

5.2.4. Enhancing medication adherence through simplified reminders

Our study found that participants struggled with complex medication regimens, often facing difficulties in remembering schedules and understanding potential side effects.

Design opportunity

How might we make medication adherence easier and more understandable for patients?

To address this, we propose an intuitive reminder system that reinforces medication adherence through a visual calendar with clear dosage instructions in simple language. Customizable reminders via push notifications or SMS ensure flexibility, while gamified elements—such as points and badges for adherence milestones—enhance engagement. This solution helps reduce missed doses, improve consistency, and strengthen patient understanding of their treatment.

5.3. Designing digital platforms to address physical recovery needs

Our findings reveal significant physical challenges faced by patients during recovery, including pain management issues and adherence to lifestyle changes. The identified Physical needs highlight several key areas where a digital solution can make a significant impact.

5.3.1. Managing post-angioplasty pain without over-reliance on painkillers

Our study found that patients experience significant pain after angioplasty, making it challenging to manage without becoming dependent on painkillers.

Design opportunity

How might we help patients manage post-angioplasty pain effectively while reducing reliance on pain medication?

To address this, we propose a digital support system for non-pharmacological pain management, offering guided relaxation techniques, breathing exercises, and physiotherapy-based movements. A personalized pain tracker enables patients to log pain levels and receive tailored relief suggestions. Educational resources on heat therapy, lifestyle adjustments, and gradual activity resumption empower patients to manage pain effectively, promoting safer recovery while reducing reliance on medication.

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5.3.2. Addressing discomfort and physical limitations after angioplasty

Our study found that many patients experience discomfort at the catheter insertion site or face physical limitations that hinder their recovery.

Design opportunity

How might we help patients manage post-procedure discomfort and overcome physical limitations for a smoother recovery?

To address this, we propose a guided recovery support system that provides educational content on proper wound care, safe movement techniques, and gradual activity resumption to help patients regain confidence in their recovery. Additionally, integrating gentle, doctor-approved rehabilitation exercises and posture guidelines can aid in preventing complications and promoting long-term well-being.

5.4. Implications for future research

The study's findings, specifically the documented informational, physical, and emotional needs of low-income angioplasty patients and their caregivers (Results file, Tables 3-5), offer several concrete implications for the design research community engaged in developing digital health interventions.

- Given the diversity of information sources impacting decision-making, design research must move beyond traditional user-centered approaches towards participatory design methods. Specifically, engaging patients and caregivers as *co-designers* is critical. Conducting co-design workshops where patients and caregivers directly contribute to the design of platform features.
- Given that patients often rely on non-medical sources for information, ("Diverse Information Sources Impact Decision-Making"), design research must focus on establishing credibility and trust within the digital intervention. This translates to co-designing the information architecture and content with patients and caregivers.
- Addressing uncertainty and fear about recovery, research should explore the effectiveness of integrating peer-to-peer support systems, moderated Q&A sessions with healthcare providers, and personalized reassurance messaging into digital health platforms.
- Recognizing the feelings of isolation and uncertainty experienced by patients and caregivers, design research should explore methods for creating a sense of community and reassurance within the digital intervention.

6. Limitations

This study offers valuable insights into the needs of angioplasty patients and their caregivers but has certain limitations. The small sample size (10 patients, 21 caregivers) may not fully capture the diversity of experiences. The demographic skew, with most patients aged 46+ and a gender imbalance (7 males, 3 females), limits applicability to younger and more diverse populations. Additionally, recovery experiences varied, with 8 patients undergoing emergency angioplasty and only 2 having elective procedures. These factors highlight the need for broader research across different demographics and healthcare settings to ensure more comprehensive and generalizable findings.

7. Conclusion

This research underscores the unique challenges low-income angioplasty patients and their caregivers' face, from financial constraints affecting medication adherence to the psychological burden of recovery. By shifting the focus from conventional healthcare interventions to human-centered design solutions, this study identifies opportunities for digital health platforms to provide accessible, contextually relevant support. Reliable health information, AI-driven symptom tracking, and interactive behavioural guidance can empower patients and caregivers, reducing misinformation and improving recovery outcomes. Future research should further investigate how inclusive design strategies can transform post-angioplasty care, making it more equitable and adaptable to diverse socioeconomic settings.

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